ic_autolab round 27 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T08:32:31Z at git `434c90658ecca530d20aad59898c585658f7108a`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-17a0be` (`python3 tools/pcc/pcc.py standing PCC-20261104-17a0be`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**improvement**	wsum2g-b30-n2	0.829 [0.761, 0.903]	0.813 [0.741, 0.892]	0.731 [0.640, 0.835]	sel < inc	2.584 / 2.711 / 3.855	3.117 / 3.333 / 5.275	1.273 / 1.388 / 2.428	2.0x / 2.0x / 1.6x
28	**improvement**	wsum2g-b60-n2	0.946 [0.895, 0.999]	0.915 [0.862, 0.971]	0.764 [0.702, 0.831]	sel < inc	2.373 / 2.483 / 3.472	2.510 / 2.713 / 4.546	0.994 / 1.087 / 1.924	2.4x / 2.3x / 1.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.8289 [0.761, 0.903] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.9456 [0.895, 0.999] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

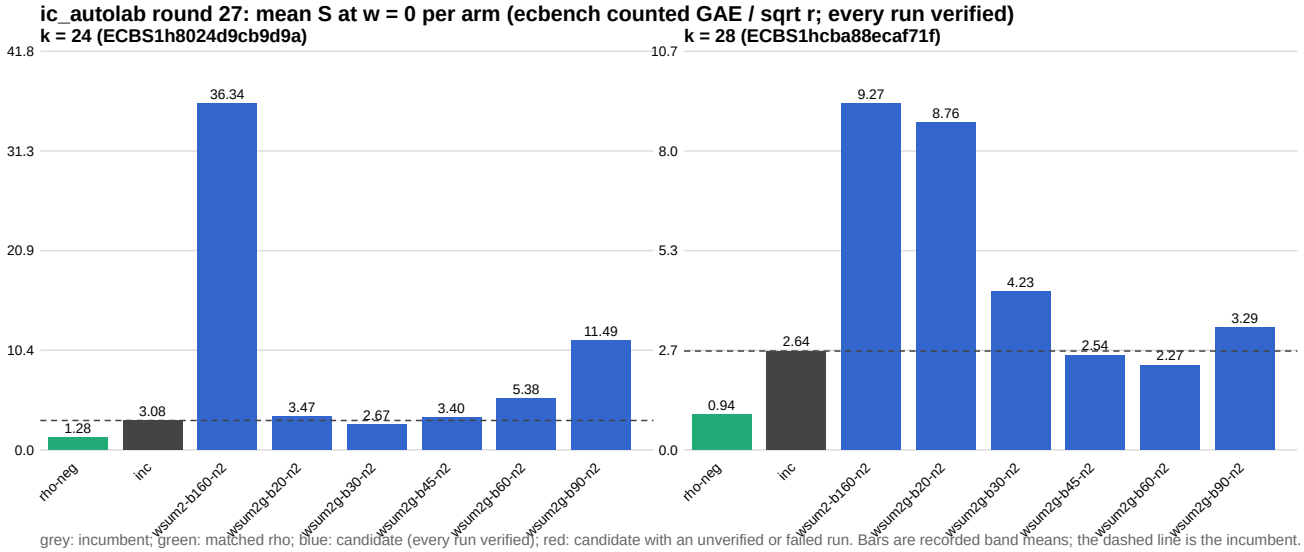
k = 24: session `ECBS1h8024d9cb9d9a` (run exit 0, verify exit 0, receipt sha256 `7c9d8f9e3c535d2d347a1d6415645f5e2beb34bda5c14576c667024fd0dbc410`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.284 / 1.401 / 2.451	matched reference
`inc`	wsum2-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	3.085 / 3.296 / 5.201	incumbent
`wsum2-b160-n2`	wsum2-b160	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=160,free=1`	32	32	36.343 / 36.353 / 36.440	exploration: family wsum2-b160 untried at k = 24, N = 0.0001 x N_a
`wsum2g-b20-n2`	wsum2g-b20	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=20,free=1`	32	32	3.466 / 3.752 / 6.323	exploration: family wsum2g-b20 untried at k = 24, N = 0.0001 x N_a
`wsum2g-b30-n2`	wsum2g-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	2.672 / 2.808 / 4.031	exploration: family wsum2g-b30 untried at k = 24, N = 0.0001 x N_a
`wsum2g-b45-n2`	wsum2g-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	3.405 / 3.456 / 3.919	exploration: family wsum2g-b45 untried at k = 24, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2g-b60-n2	wsum2g-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	5.376 / 5.405 / 5.667	exploration: family wsum2g-b60 untried at k = 24, N = 0.0001 x N_a
wsum2g-b90-n2	wsum2g-b90	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	11.491 / 11.506 / 11.642	exploration: family wsum2g-b90 untried at k = 24, N = 0.0001 x N_a

k = 28: session `ECBS1hcba88ecaf71f` (run exit 0, verify exit 0, receipt sha256 `efd299263878b247ad9c2db1640373b665fd7066b1bd1b11dedfa179b170480d`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.938 / 1.026 / 1.812	matched reference
inc	wsum2-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.644 / 2.874 / 4.946	incumbent
wsum2-b160-n2	wsum2-b160	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=160,free=1	32	32	9.274 / 9.306 / 9.595	exploration: family wsum2-b160 untried at k = 28, N = 0.0001 x N_a
wsum2g-b20-n2	wsum2g-b20	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=20,free=1	32	32	8.765 / 9.626 / 17.375	exploration: family wsum2g-b20 untried at k = 28, N = 0.0001 x N_a
wsum2g-b30-n2	wsum2g-b30	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	4.232 / 4.622 / 8.133	exploration: family wsum2g-b30 untried at k = 28, N = 0.0001 x N_a
wsum2g-b45-n2	wsum2g-b45	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=45,free=1	32	32	2.540 / 2.722 / 4.355	exploration: family wsum2g-b45 untried at k = 28, N = 0.0001 x N_a
wsum2g-b60-n2	wsum2g-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.274 / 2.374 / 3.273	exploration: family wsum2g-b60 untried at k = 28, N = 0.0001 x N_a
wsum2g-b90-n2	wsum2g-b90	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	3.286 / 3.331 / 3.734	exploration: family wsum2g-b90 untried at k = 28, N = 0.0001 x N_a



train screen

Selection

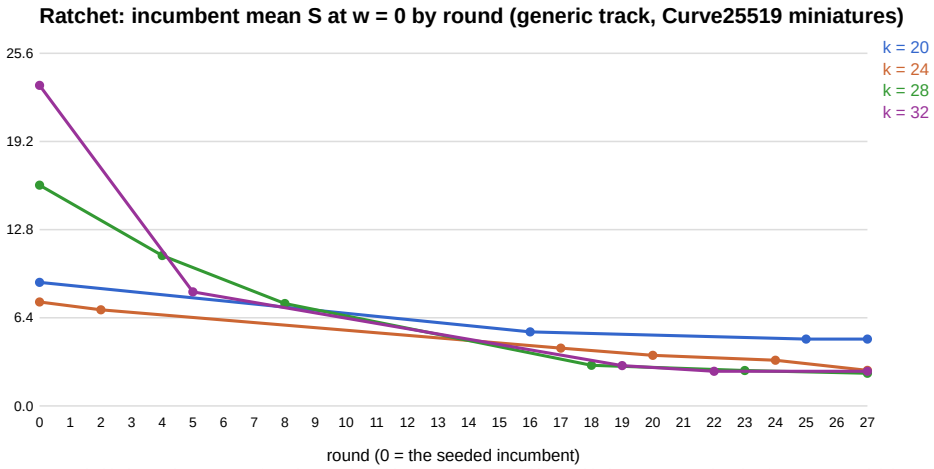
- k = 24: challenger `wsum2g-b30-n2` (Pareto alternative at w = 1: `wsum2g-b45-n2`)
- k = 28: challenger `wsum2g-b60-n2` (Pareto alternative at w = 1: `wsum2g-b60-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
24	`ECBS1h30a8186f15ae`	0	0	`327358af12478f683557713cbb1329fced6729d3a01134d3e7af7016cd901f3c`	{“verified”:480}
28	`ECBS1h77f9661c8891`	0	0	`1a18510be95320ce57bf410ef1be28b347eae495d52e7ab9efa4ee54e2334e2a`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	25	4.860 / 5.083 / 7.092	3.2x / 3.2x / 2.8x
24	`wsum2g-b30-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=30,free=1`)	27	2.584 / 2.711 / 3.855	2.0x / 2.0x / 1.6x
28	`wsum2g-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	27	2.373 / 2.483 / 3.472	2.4x / 2.3x / 1.8x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `6229f85b81f9a2e8f2811d675f77f4f830ca536422a1e92680a382183450a62c`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.